



Let's walk through the most **common system design tools** We have learnt, explain **what they do**, and **when to use each one**.

OVERVIEW TABLE

Tool	Purpose	When to Use
DFD (Data Flow Diagram)	Shows how data flows in a system	When analyzing processes & data movement
System Flowchart	Shows physical or logical flow of information and actions	When mapping system operations or hardware/software interaction
Decision Table	Lists all possible conditions and actions	When rules or decisions are complex
HIPO Chart	Breaks system into modules and shows their inputs/ outputs	During planning and modular design
ER Diagram (Entity-Relationship)	Describes how data entities are related in a database	When designing the database structure

DETAILED BREAKDOWN

1. DFD – Data Flow Diagram

- **What it shows:**
How data moves through the system, between users, processes, and storage.

- **Best For:**

Understanding the **functional flow** of data, especially in early system analysis.

- **When to use:**

- During requirement gathering
 - When showing how data is input, processed, and stored
 - When you want a **logical view** of the system
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2. ↻ System Flowchart

- **What it shows:**

The **sequence** of operations in a system — including **decisions**, **data inputs**, and **outputs**.

- **Best For:**

Explaining **how the system runs** (especially in legacy or detailed technical diagrams).

- **When to use:**

- To document system **logic**
 - For **hardware/software interaction** diagrams
 - During **implementation** or when improving existing processes
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3. ⚡ Decision Table

- **What it shows:**

A matrix of all possible **conditions and corresponding actions**.

- **Best For:**

Business rules or program logic with **lots of combinations**.

- **When to use:**

- When decisions are based on **multiple conditions**
 - In **testing**, validation, and rules automation
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4. 🏗️ HIPO Chart (Hierarchy + IPO)

- **What it shows:**

Program is broken into **modules**, each showing its **Input, Process, Output**.

- **Best For:**

Planning **structured programs** or systems.

- **When to use:**

- Early in design phase
 - To **break a big system** into manageable pieces
 - When documenting or explaining **modular logic**
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5. ➡️ ER Diagram – Entity-Relationship Diagram

- **What it shows:**

Entities (tables), their **attributes**, and **relationships**.

- **Best For:**

Designing and understanding a **relational database**.

- **When to use:**

- During **database design**
 - When analyzing relationships between data types
 - To prepare for implementation in SQL or NoSQL systems
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✓ FINAL RECOMMENDATION — When to Use What:

Phase	Best Tools
Requirement Gathering	DFD, System Flowchart
System Analysis	DFD, Decision Table, ER Diagram
Design Phase	HIPO Chart, ER Diagram, System Flowchart

Phase	Best Tools
Implementation Planning	HIPO Chart, Decision Table, Flowchart
Database Design	ER Diagram
Testing & Validation	Decision Table, DFD

◆ Summary Quote for Students:

"**DFD** shows how data moves,
ER Diagram shows how data is structured,
Flowcharts show how the system runs,
Decision tables show rule logic,
HIPO shows what each module does."
